

IOWA STATE UNIVERSITY
ELECTRICAL AND COMPUTER ENGINEERING

EE 230
LAB 8 REPORT

April 21, 2026

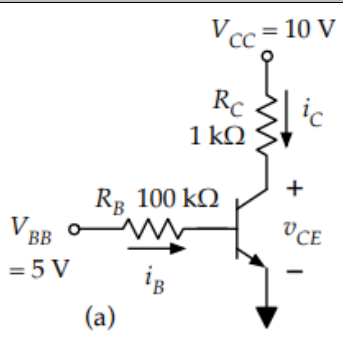
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Introduction

In this lab, we started working on BJT transistors. At first, we took 4 different transistors, including the PN2222 and PN2907 transistors, and used them in the parameter analyzer to find their β_F values and get the i_C - V_{CE} traces. We compared the value with our measured one and recorded them for use in the next experiments. Then we built 3 different circuits with the PN2222 and one with the PN2907, and measured the base, collector and emitter current along with the base-emitter voltage and collector-emitter voltage. Then we compared those values with our calculated values. Then we made a BJT switch with the PN2222 and collected data across 25 different V_i and visualized the behavior through graphs. Finally, we built an amplifier circuit with a PN2222 to light up a LED and observed how adding the BJT improved the performance, and then we proceeded to build an amplifier with a BJT.

A Quick measure of β_F and $i_C - v_{CE}$ characteristics with the parameter analyzer

Circuits	Calculated Values
 (a)	Collector Current, $i_C = 9.2 \text{ mA}$ Base Current, $i_B = 0.043 \text{ mA}$ Forward current gain, $\beta_F = 213.95$ $\beta_F = i_C/i_B = 213.95$

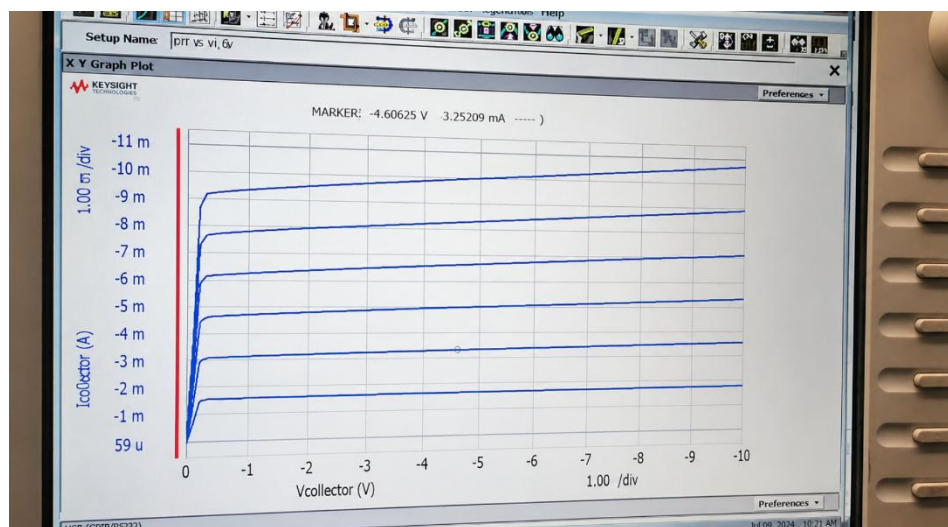
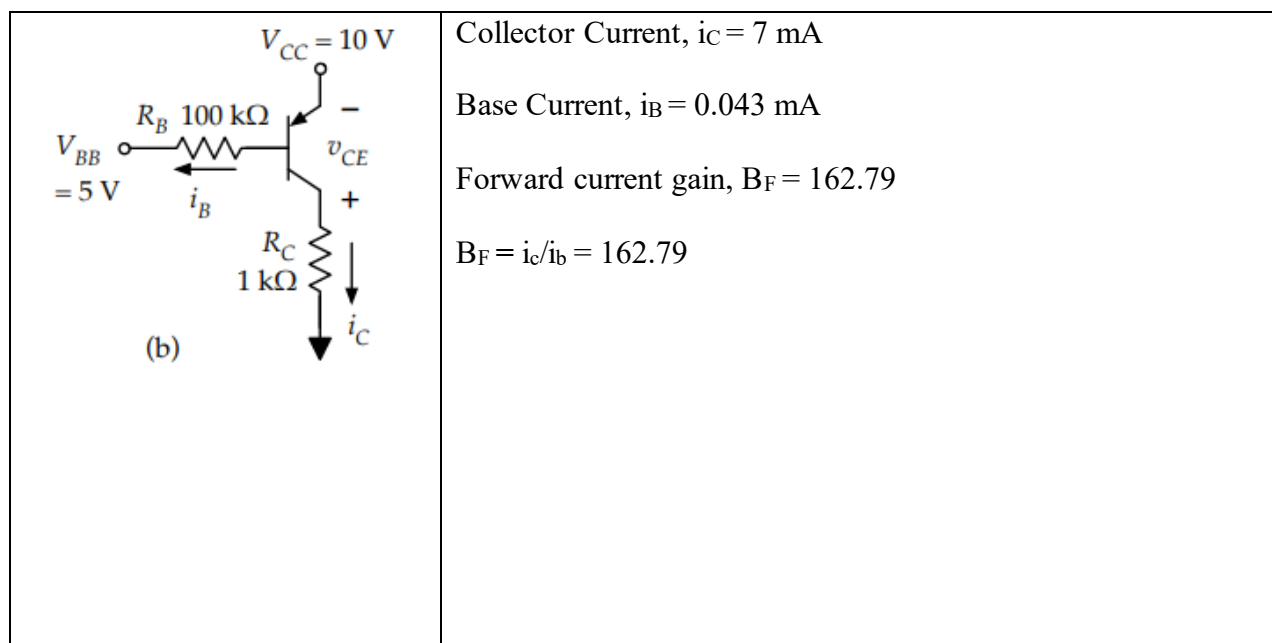


Figure 1: $i_C - v_{CE}$ curves for PN2222

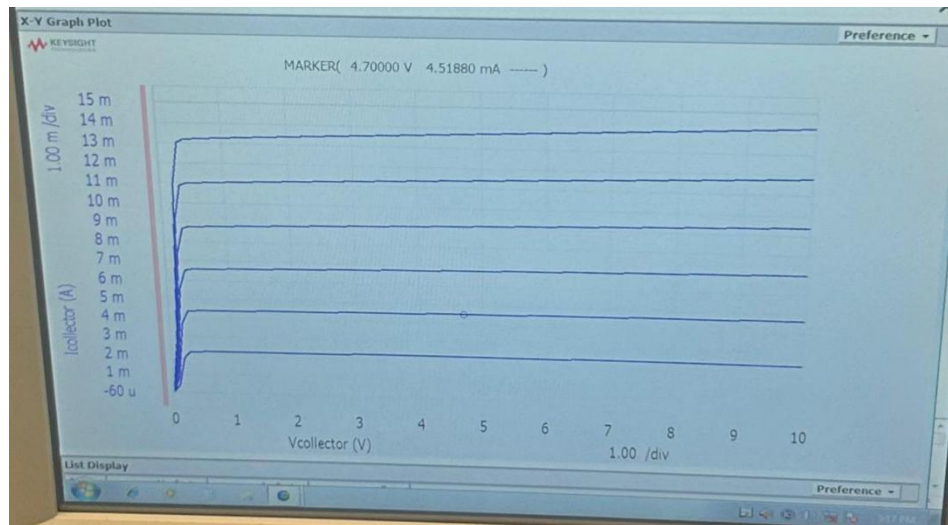
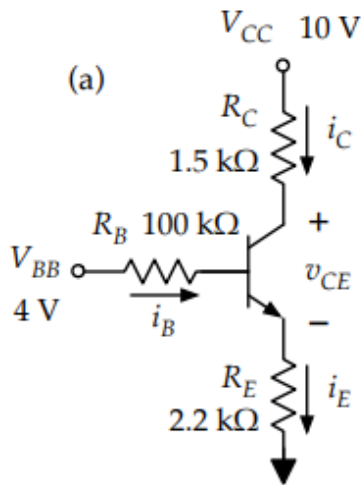


Figure 2: $i_C - v_{CE}$ curves for PN2906

B Simple BJT Circuits



	i_B	i_C	i_E	V_{BE}	V_{CE}
Measured	0.0055 mA	1.27 mA	1.82 mA	0.63 V	5.28 V
Calculated	0.00576 mA	1.232 mA	1.238 mA	0.7 V	5.43 V

a) assuming,

$$V_{BE} = 0.7 \text{ V}$$

we know,

$$i_E = i_B + i_C$$

$$\Rightarrow i_E = i_B + \beta i_B$$

$$\Rightarrow i_E = (\beta + 1) i_B$$

also, using KVL on the base-emitter loop,

$$V_{BB} - i_B R_B - V_{BE} - i_E R_E = 0$$

$$\Rightarrow V_{BB} = \left(\frac{i_E}{\beta + 1} \right) R_B + V_{BE} + i_E R_E$$

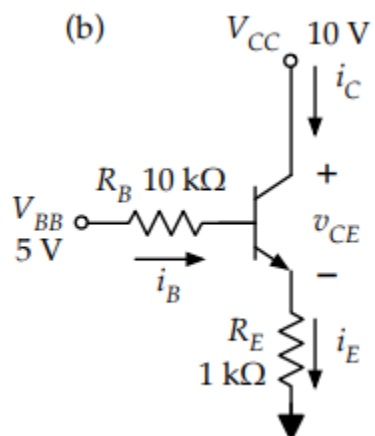
$$\Rightarrow i_E = \frac{V_{BB} - V_{BE}}{R_E + \frac{R_B}{\beta + 1}}$$

$$\Rightarrow i_E = 1.238 \text{ mA}$$

$$i_B = \frac{i_E}{\beta + 1} = \frac{1.238 \text{ mA}}{1 + 213.95} = 0.00576 \text{ mA}$$

$$i_C = \beta i_B = 1.232 \text{ mA}$$

$$V_{CE} = V_{CC} - i_C R_C - i_E R_E = 5.43 \text{ V}$$



	i_B	i_C	i_E	V_{BE}	V_{CE}
Measured	0.018 mA	4.082 mA	4.16 mA	0.66 V	5.83 V
Calculated	0.0191 mA	4.090 mA	4.109 mA	0.7 V	5.89 V

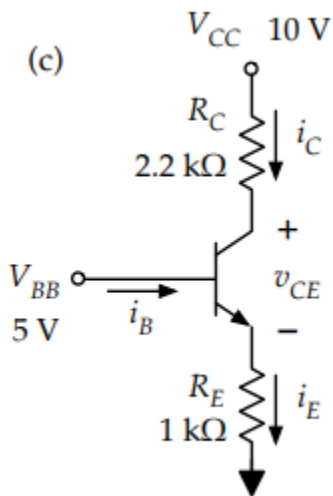
$$b) \quad i_E = \frac{V_{BB} - V_{BE}}{R_E + \frac{R_B}{1+\beta}} = 4.109 \text{ mA}$$

$$i_B = \frac{i_E}{1+\beta} = \frac{4.109 \text{ mA}}{1+213.95} = 0.0191 \text{ mA}$$

$$i_C = \beta i_B = (213.95)(0.0191) = 4.090 \text{ mA}$$

also,

$$V_{CE} = V_C - V_E = 10 - i_E R_E = 5.89 \text{ V}$$



	i_B	i_C	i_E	V_{BE}	V_{CE}
Measured	1.68 mA	2.6 mA	4.28 mA	0.726	0.096 V
Calculated	1.80 mA	2.04 mA	4.3 mA	0.7 V	0.2 V

d) $V_B = 5V$ $V_{BE} = 0.7V$

$$V_E = V_B - V_{BE} = 4.3V$$

$$i_E = \frac{V_E}{R_E} = 4.3 \text{ mA}$$

assuming, the transistor is saturating,

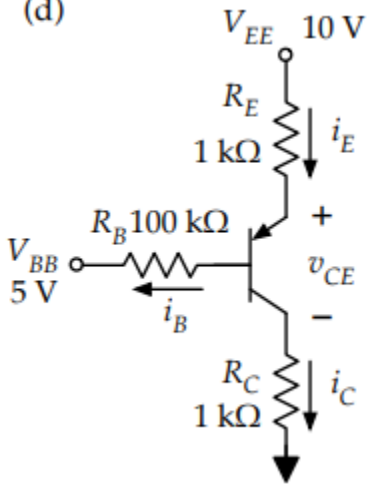
$$V_{CE} \approx 0.2V$$

$$V_C = V_{CE} + V_E = 4.5V$$

$$i_C = \frac{V_C}{R_C} = \frac{4.5}{2.2k} = 2.04 \text{ mA}$$

$$i_B = i_E - i_C = 1.80 \text{ mA}$$

(d)



	i_B	i_C	i_E	V_{BE}	V_{CE}
Measured	0.016 mA	2.7 mA	2.7 mA	-0.66 V	-4.59 V
Calculated	0.0163 mA	2.65 mA	2.67 mA	-0.7 V	-4.68 V

d) $\beta = 162.79$

using KVL,

$$i_E = (\beta + 1)i_B$$

— (i)

$$10 - i_E R_E - 0.7 - i_B R_B - 5 = 0$$

— (ii)

using (i) & (ii) $\Rightarrow$

$$4.3 = i_B(100k) + i_B(163.79)(1k)$$

$$\Rightarrow i_B = 0.0163 \text{ mA}$$

$$i_C = \beta i_B = 2.68 \text{ mA}$$

$$i_E = i_C + i_B = 2.67 \text{ mA}$$

now,

$$V_E = 10 - i_E R_E = 7.33 \text{ V}$$

$$V_C = i_C R_C = 2.68 \text{ V}$$

$$V_{CE} = V_C - V_E = -4.68 \text{ V}$$

$$V_{BE} = -0.7 \text{ V}$$

C BJT as a switch

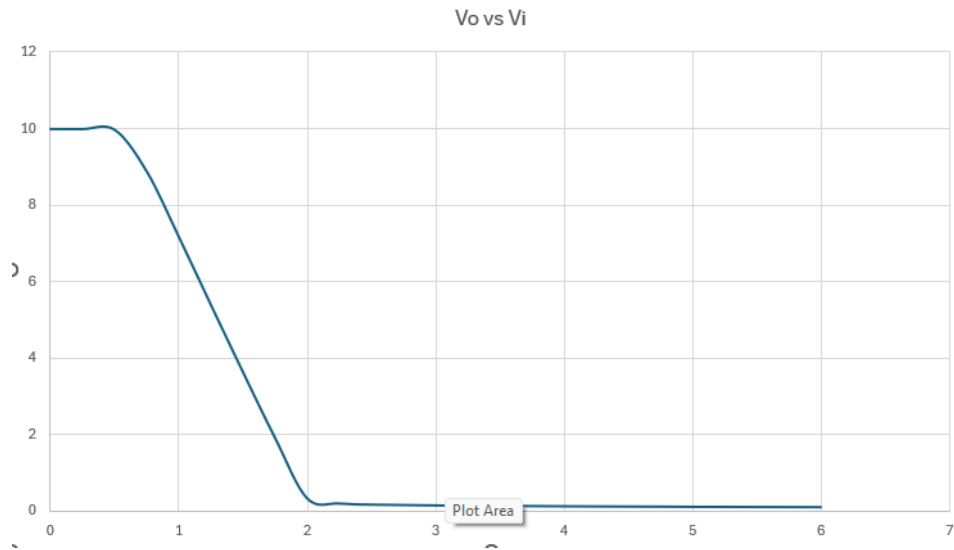


Figure 4: Transfer Characteristics, plot of v_o vs v_i

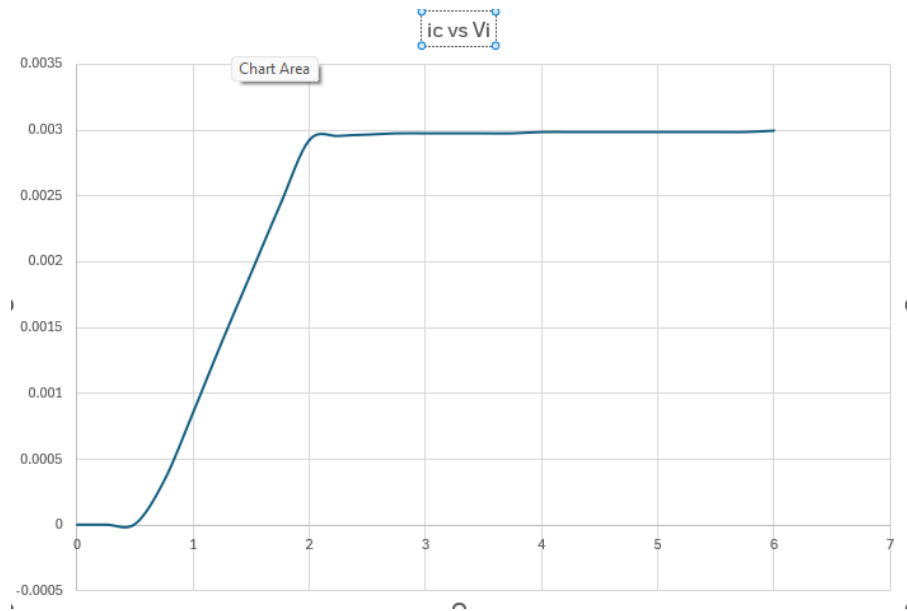


Figure 5: Plot of i_c vs v_i

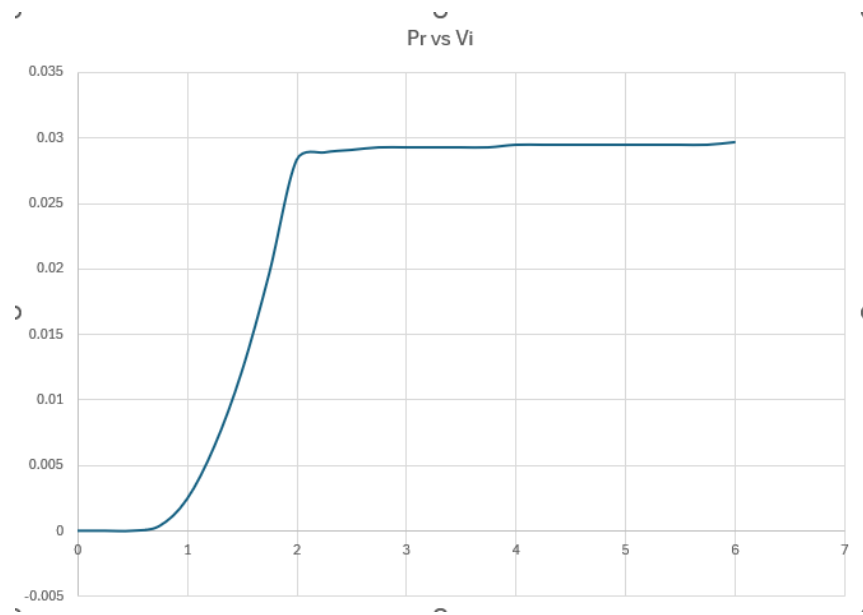


Figure 6: Plot of P_R vs v_i

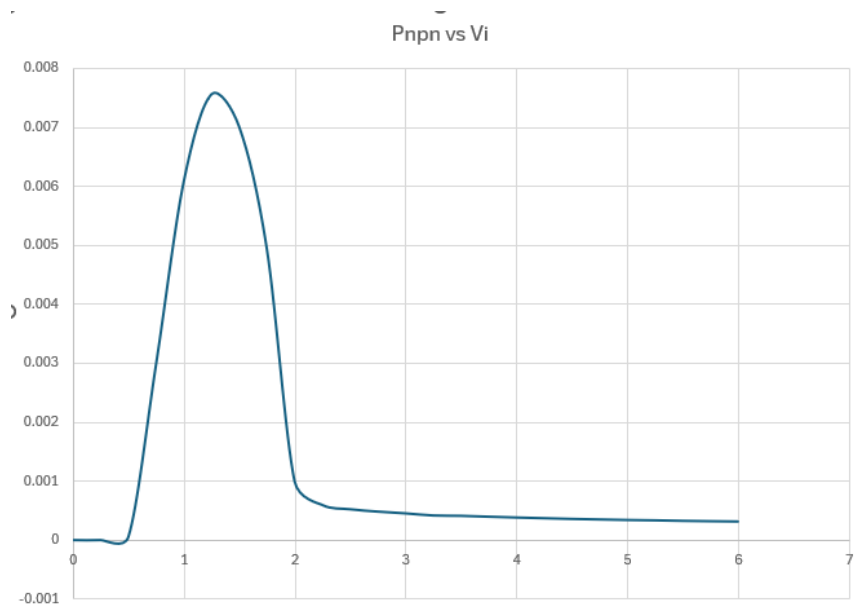


Figure 7: Plot of P_{npn} vs v_i

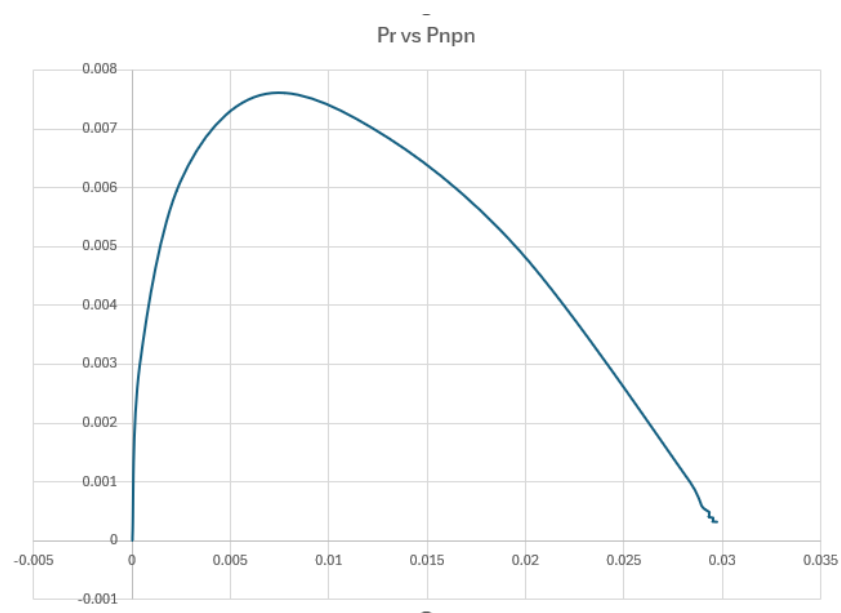
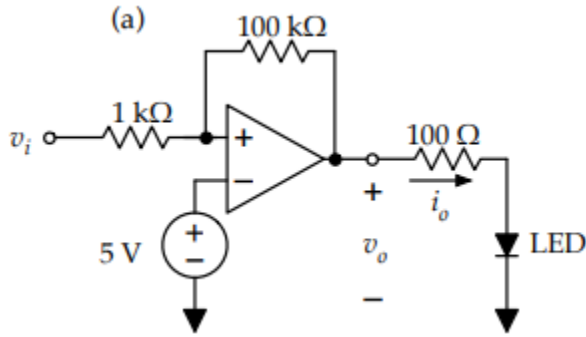
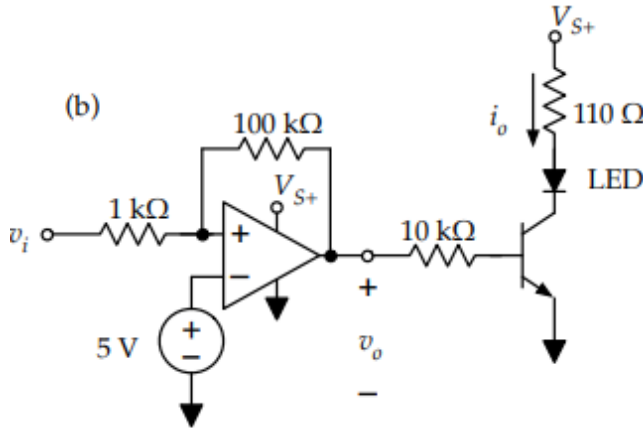


Figure 8: Plot of P_R vs P_{npn}



Measured Output Voltage, $v_o = 7.57 \text{ V}$

Measured Output Current, $i_o = 0.757 \text{ mA}$



Measured Output Voltage, $v_o = 9.95 \text{ V}$

Measured Output Current, $i_o = 72.7 \text{ mA}$

In part a, the voltage and current are lower than ideal because the op-amp is directly driving the LED load. The amp cannot source a large output current. As such when the LED and resistor demand current, the output voltage drops below the 10 V supply rail. But in part b, the transistor output stage fixes this problem by letting the op amp supply only a small base current while the transistor supplies the larger LED current.

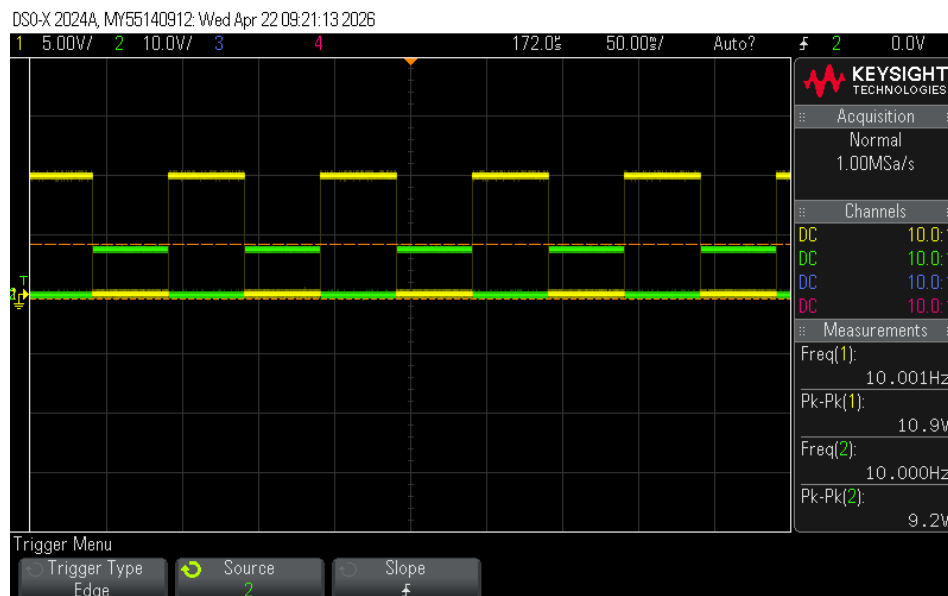


Figure 9: Oscilloscope Trace

D BJT as an amplifier

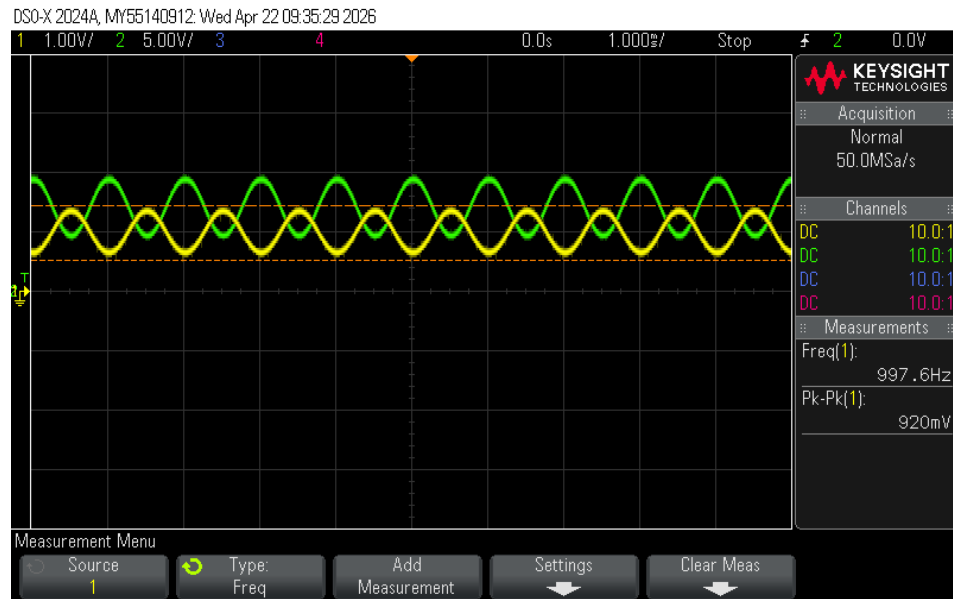


Figure 10: Oscilloscope Trace

Gain = 6.72

Conclusion

In this lab, we didn't face any major problems. All the experiments were completed without any issues, and all our measured values were very close to our calculated values, they were all within 5% of our calculated values. The plots we derived from excel also matched our expectations. The lab helped us visualize the theory we learned about BJTs.